

B.Tech. Degree VI Semester Special Supplementary Examination November 2012

ME 603 MACHINE DESIGN I
(2006 Scheme)

Time : 3 Hours

Maximum Marks : 100

(Design Data Hand book permitted)

PART A
(Answer *ALL* questions)

(8 × 5 = 40)

- I. (a) List out six factors which governs the design of machine part. Briefly explain any two.
(b) State and explain the maximum normal stress theory.
(c) For transmitting power, square threads are preferable to V-threads. Why?
(d) Design a sunk key considering shearing strength and crushing strength.
(e) What is the difference between caulking and fullering? Explain with the help of neat sketches.
(f) Classify the springs and name the principal stress induced in each.
(g) Design a double transverse fillet weld under tension.
(h) A hollow shaft has greater strength and stiffness than a solid shaft of equal weight. Explain.

PART B

(4 × 15 = 60)

- II. (a) State and explain the maximum shear stress theory. (5)
(b) A bolt is subjected to an axial pull of 10kN and a transverse shear force of 5kN. The yield strength of the bolt material is 300MPa. Considering a factor of safety 2.5, determine the diameter of the bolt, using (i) maximum normal stress theory (10)
(ii) maximum shear stress theory (iii) maximum principal strain theory. Take Poisson's ratio as 0.25.

OR

- III. (a) What are the methods proposed to calculate the safe values of fluctuating stresses? Explain any one. (5)
(b) A rod of circular cross-section is subjected to an alternating tensile force, varying from 20kN to 70kN. Determine the diameter of the rod, according to (10)
(i) Gerber methods
(ii) Goodman method
(iii) Soderberg method,

using the following material properties:

Ultimate tensile strength = 1000 MPa.
Yield strength = 550 MPa.

Take factor of safety as 2. Neglect stress concentration effect and other correction factors.

(P.T.O)

- IV. (a) Derive the expressions for torque required to raise and to lower the load in a screw jack of square threads. (7)
- (b) The following data relate to a screw jack:

Load	= 20 kN
Nominal diameter of the screw	= 40mm
Pitch of the screw	= 6mm
Coefficient of friction between the screw and nut	= 0.15

Assuming that the load rotates with the screw, determine (i) the torque required to raise and to lower the load (ii) the efficiency of the screw jack. (8)

OR

- V. A mild steel shaft has to transmit 70KW at 240 rpm. The allowable shear stress in the shaft material is limited to 45 MPa, and the angle of twist is not to exceed 1° in a length of 20 times the shaft diameter. Determine the shaft diameter and design a cast iron flange coupling of protected type for the shaft. The shear stress in the coupling bolts is to be limited to 30 MPa. (15)

Permissible shear stress for C.I. flange = 15 MPa

Shaft and key are made of the same material.

Permissible crushing stress for key material = 100MPa.

Modulus of rigidity of shaft material = 0.84×10^5 MPa.

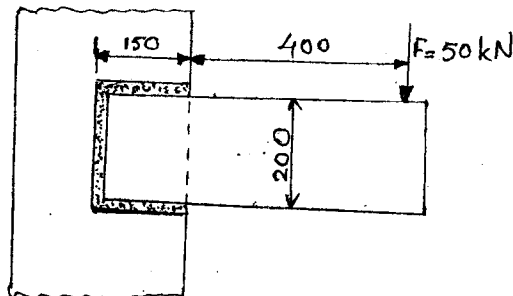
Draw a suitable neat diagram of the coupling.

- VI. (a) Briefly explain any two mode of failure of riveted joint. (5)
- (b) Design the longitudinal joint for a 1.25m diameter steam boiler to carry a steam pressure of 2.5N/mm^2 . The ultimate strength of the boiler plate may be assumed as 420MPa, crushing strength as 650 MPa and shear strength as 300MPa. Take the rivet joint efficiency as 80%. Sketch the joint with all the dimensions. Take suitable factor of safety as 5. (10)

OR

- VII. (a) Explain nipping of a leaf spring. (5)
- (b) A bumper, consisting of two helical steel springs of circular section, bridge to rest a railway wagon of mass 1500Kg, and moving at 1.2 m/s. While doing so, the springs are compressed by 150mm. The mean diameter of the coils is 6 times the wire diameter. The permissible shear stress is 400MPa. Determine (i) maximum force on each spring (ii) wire diameter of the spring (iii) mean diameter of the coils (iv) number of active coils. Take $G = 0.84 \times 10^5$ MPa. (10)

- VIII. Figure shows an eccentrically loaded welded joint. Determine the fillet weld size. (15)
- Allowable shear stress in the weld is 80MPa.



OR

- IX. A shaft is required to transmit 15KW at 200rpm. It is supported on two bearings, 0.6m apart and two gears are keyed on it. A 18 teeth, $14\frac{1}{2}^\circ$ involute 5mm module pinion is located at 120mm to the right of the right bearing, and delivers power to a gear directly below the shaft. A 72 tooth, $14\frac{1}{2}^\circ$ involute, 5mm module gear is located at 150mm to the right of the left bearing, and receives power from a gear, directly above it. Determine the shaft diameter, assuming permissible normal stress as 80MPa. (15)